Full Stack Development — Mini E-Commerce Project
Lab Sheet 2: Making It Interactive (Vanilla JavaScript)

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```
const products = [
  {
    id: 1,
    name: "Wireless Headphones",
    price: 1999,
    image: "MyStore-Images/wireless-headphones.jpg"
  },
  {
    id: 2,
    name: "Smart Watch",
    price: 2499,
    image: "MyStore-Images/SmartWatch.jpg"
  },
  {
    id: 3,
    name: "Laptop Backpack",
    price: 1299,
    image: "MyStore-Images/laptop-backpack.jpg"
  },
  {
    id: 4,
    name: "Mechanical Keyboard",
    price: 2999,
    image: "MyStore-Images/mechanical-keyboard.jpg"
  },
  {
    id: 5,
    name: "Gaming Mouse",
    price: 999,
    image: "MyStore-Images/gaming-mouse.jpg"
  },
  {
    id: 6,
    name: "USB-C Hub",
    price: 1499,
    image: "MyStore-Images/usb-c-hub.jpg"
  },
  {
    id: 7,
    name: "Bluetooth Speaker",
    price: 1799,
```

```
        image: "MyStore-Images/bluetooth-speaker.jpg"
    },
    {
        id: 8,
        name: "HD Webcam",
        price: 1599,
        image: "MyStore-Images/webcam.jpg"
    }
];


let cart = JSON.parse(localStorage.getItem("cart")) || [];


function updateCartCount() {

    const cartCount = document.querySelector("#cart-count");

    if (cartCount) {

        let totalItems = 0;

        cart.forEach(function(item) {
            totalItems += item.quantity || 1;
        });

        cartCount.textContent = totalItems;
    }
}


const productContainer = document.querySelector("#product-container");

if (productContainer) {

    products.forEach(function(product) {

        const productCard = document.createElement("div");

        productCard.classList.add("product-card");

        productCard.innerHTML = `
            <a href="product-detail.html?id=${product.id}">
                <img src="${product.image}" alt="${product.name}">
            </a>

            <h3>
                <a href="product-detail.html?id=${product.id}">
                    ${product.name}
                </a>
```

```
        </h3>

        <p>₹${product.price}</p>

        <button class="add-to-cart" data-id="${product.id}">
            Add to Cart
        </button>
    `;

    productContainer.appendChild(productCard);
});


const addToCartButtons =
    document.querySelectorAll(".add-to-cart");

addToCartButtons.forEach(function(button) {

    button.addEventListener("click", function() {

        const productId = Number(button.dataset.id);

        const product = products.find(function(product) {
            return product.id === productId;
        });

        const existingItem = cart.find(function(item) {
            return item.id === productId;
        });

        if (existingItem) {

            existingItem.quantity += 1;

        } else {

            cart.push({
                ...product,
                quantity: 1
            });
        }

        localStorage.setItem("cart", JSON.stringify(cart));

        updateCartCount();

        alert(product.name + " added to cart!");

    });
});
```

```
}


const productDetail = document.querySelector("#product-detail");

if (productDetail) {

    const urlParams =
        new URLSearchParams(window.location.search);

    const productId =
        Number(urlParams.get("id"));

    const product =
        products.find(function(product) {
            return product.id === productId;
        });

    if (product) {

        productDetail.innerHTML = `
            <img src="${product.image}" alt="${product.name}">

            <div>
                <h2>${product.name}</h2>

                <p>
                    This is a high-quality ${product.name}.
                    Perfect for everyday use.
                </p>

                <h3>₹${product.price}</h3>

                <label for="detail-quantity">
                    Quantity:
                </label>

                <input
                    type="number"
                    id="detail-quantity"
                    value="1"
                    min="1"
                >

                <br><br>

                <button id="detail-add-cart">
                    Add to Cart
                </button>
            </div>
```

```
        `;


        document
            .querySelector("#detail-add-cart")
            .addEventListener("click", function() {

                const quantity =
                    Number(
                        document.querySelector("#detail-quantity").value
                    );

                const existingItem =
                    cart.find(function(item) {
                        return item.id === product.id;
                    });

                if (existingItem) {

                    existingItem.quantity += quantity;

                } else {

                    cart.push({
                        ...product,
                        quantity: quantity
                    });
                }

                localStorage.setItem(
                    "cart",
                    JSON.stringify(cart)
                );

                updateCartCount();

                alert(product.name + " added to cart!");
            });

    } else {

        productDetail.innerHTML = `
            <h2>Product not found</h2>
        `;
    }
}


const cartContainer =
    document.querySelector("#cart-container");
```

```
if (cartContainer) {

    function renderCart() {

        cartContainer.innerHTML = "";

        if (cart.length === 0) {

            cartContainer.innerHTML = `
                <p>Your cart is empty.</p>
            `;

            calculateTotal();

            return;
        }

        cart.forEach(function(item) {

            const row =
                document.createElement("div");

            row.classList.add("cart-item");

            row.innerHTML = `
                <span>
                    ${item.name}
                </span>

                <span>
                    ₹${item.price}
                </span>

                <input
                    type="number"
                    class="cart-quantity"
                    data-id="${item.id}"
                    value="${item.quantity}"
                    min="1"
                >

                <span>
                    ₹${item.price * item.quantity}
                </span>

                <button
                    class="remove-item"
                    data-id="${item.id}">
                    Remove
```

```
        </button>
    `;

    cartContainer.appendChild(row);
});


const quantityInputs =
    document.querySelectorAll(".cart-quantity");

quantityInputs.forEach(function(input) {

    input.addEventListener("change", function() {

        const id =
            Number(input.dataset.id);

        const newQuantity =
            Number(input.value);

        const item =
            cart.find(function(item) {
                return item.id === id;
            });

        if (item && newQuantity > 0) {

            item.quantity = newQuantity;

            localStorage.setItem(
                "cart",
                JSON.stringify(cart)
            );

            renderCart();
            updateCartCount();
        }
    });
});

const removeButtons =
    document.querySelectorAll(".remove-item");

removeButtons.forEach(function(button) {

    button.addEventListener("click", function() {

        const id =
            Number(button.dataset.id);
```

```
            cart =
                cart.filter(function(item) {
                    return item.id !== id;
                });

            localStorage.setItem(
                "cart",
                JSON.stringify(cart)
            );

            renderCart();
            updateCartCount();
        });
    });

    calculateTotal();
}

function calculateTotal() {

    const totalElement =
        document.querySelector("#cart-total");

    if (!totalElement) {
        return;
    }

    const total =
        cart.reduce(function(sum, item) {

            return sum + (item.price * item.quantity);

        }, 0);

    totalElement.textContent =
        "Grand Total: ₹" + total;
}

renderCart();
}

const checkoutForm =
    document.querySelector("#checkout-form");

if (checkoutForm) {

    checkoutForm.addEventListener("submit", function(event) {

        event.preventDefault();
```

```
// Get values

const name =
   document.querySelector("#name").value.trim();

const address =
   document.querySelector("#address").value.trim();

const city =
   document.querySelector("#city").value.trim();

const pincode =
   document.querySelector("#pincode").value.trim();

const phone =
   document.querySelector("#phone").value.trim();

const nameError =
   document.querySelector("#name-error");

const addressError =
   document.querySelector("#address-error");

const pincodeError =
   document.querySelector("#pincode-error");

const phoneError =
   document.querySelector("#phone-error");

const confirmation =
   document.querySelector("#confirmation");

nameError.textContent = "";
addressError.textContent = "";
pincodeError.textContent = "";
phoneError.textContent = "";
confirmation.textContent = "";

let valid = true;

if (name === "") {

   nameError.textContent =
      "Name is required.";

   valid = false;
}

if (address === "") {
```

```
            addressError.textContent =
                "Address is required.";

            valid = false;
        }

        if (!/^\d{6}$/.test(pincode)) {

            pincodeError.textContent =
                "Pincode must be exactly 6 digits.";

            valid = false;
        }

        if (!/^\d{10}$/.test(phone)) {

            phoneError.textContent =
                "Phone must be exactly 10 digits.";

            valid = false;
        }

        if (valid) {

            confirmation.textContent =
                "Order placed successfully!";

            localStorage.removeItem("cart");

            cart = [];

            updateCartCount();

            checkoutForm.reset();
        }

    });
}

updateCartCount();
```